```
clc;
clear;
close all;
% (7,4) Hamming Code
msg = [1 0 1 1];
% Encode
code = encode(msg,7,4,'hamming/binary');
% Introduce Single-Bit Error
rx = code;
rx(3) = ~rx(3); % Flip one bit
% Decode and Correct Error
dec = decode(rx,7,4,'hamming/binary');
% Display Results
disp('Message Bits:')
```

```
Message Bits:
```

```
disp(msg)
```

```
     1     0     1     1
```

```
disp('Encoded Bits:')
```

```
Encoded Bits:
```

```
disp(code)
```

```
     1     0     0     1     0     1     1
```

```
disp('Received Bits (with error):')
```

```
Received Bits (with error):
```

```
disp(rx)
```

```
     1     0     1     1     0     1     1
```

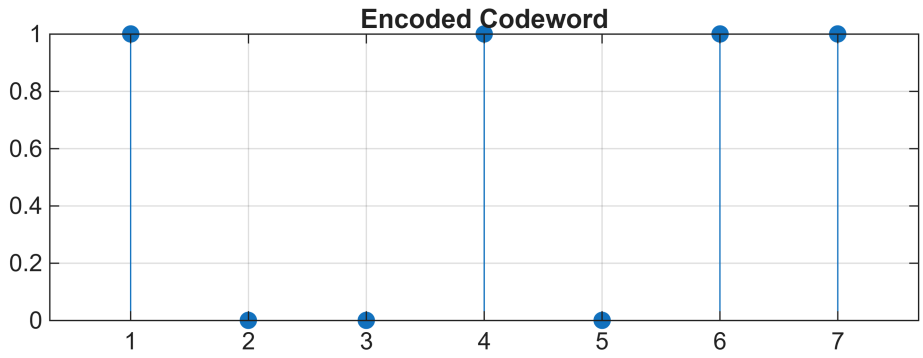
```
disp('Corrected Message:')
```

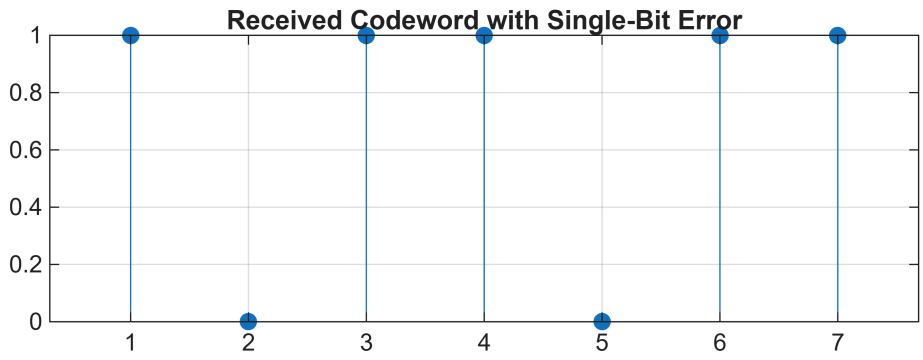
```
Corrected Message:
```

```
disp(dec)
```

```
     1     0     1     1
```

```
% Plot
figure;
subplot(2,1,1)
stem(code,'filled')
title('Encoded Codeword')
grid on
subplot(2,1,2)
stem(rx,'filled')
title('Received Codeword with Single-Bit Error')
grid on
```


Encoded Codeword
1
0.8
0.6
0.4
0.2
0
1
2
3
4
5
6
7


Received Codeword with Single-Bit Error
1
0.8
0.6
0.4
0.2
0
1
2
3
4
5
6
7